

Exposing the Long-tail in Embodied Urban Navigation via Scalable Learning from In-the-Wild Videos

Supplementary Material

Anonymous submission

A Setup of WILD-LongTail.

Dataset Partition

WILD-LongTail is constructed from our large-scale in-the-wild video collection as a diagnostic benchmark for systematically characterizing the long-tail distribution of urban navigation. It is disjoint from the conventional video-learning splits at the source-video level and comprises two components: a Long-Tail Mining Set containing 287,820 video-derived navigation episodes and a held-out Long-Tail Test Set containing 104,655 episodes with fixed identifiers. The remaining video episodes are used as additional training data for policy learning or model adaptation. The Long-Tail Mining Set is used to estimate perception-motion feature frequencies, identify distributionally rare patterns and model-dependent high-error cases, conduct reflection-based failure analysis, and construct the incremental fine-tuning subsets. Specifically, +Random contains 15K randomly sampled mining episodes, +Hard contains 7.5K mined hard episodes, and +Hard+Random combines the same 7.5K hard episodes with 7.5K randomly sampled replay episodes, yielding a total of 15K episodes. All incremental variants are fine-tuned for one epoch. The Long-Tail Test Set is reserved exclusively for evaluation: Test-all denotes the complete set of 104,655 held-out episodes, whereas Test-hard is a fixed subset of Test-all selected once using the frozen WILD-Nav-inst model before any incremental fine-tuning. Test-hard is defined as the union of episodes ranked by the ADE and FDE error. The resulting episode identifiers remain unchanged across all subsequent evaluations, ensuring that every model is compared on exactly the same difficult cases. Therefore, Test-hard is an evaluation subset of Test-all rather than an additional independent split, and no episode from the Long-Tail Test Set is used for fine-tuning or hyperparameter selection.

Trajectory Reconstruction

The reconstruction is performed at 3 Hz and a 32-second temporal window to preserve short-term motion dynamics. We define 20 meta-actions to discretize continuous future

trajectories into interpretable navigation primitives. Each trajectory is transformed into an ego-centric coordinate system centered at the current frame and represented by five future waypoints in forward-left coordinates. The taxonomy was informed by exploratory trajectory clustering but uses deterministic geometric rules for stable annotation. Valid trajectories are then categorized as stop, backward, or moving actions, where stop requires less than 1.2 m net displacement and 1.8 m path length. Moving trajectories are divided into slow, normal, and fast tiers using mean-step thresholds of 0.75 and 1.05 m and path-length thresholds of 3.75 and 5.25 m. And they are further classified by geometry as straight motion, left/right-biased forward motion, left/right turns, or lateral translations. Overall, the taxonomy is dominated by speed-aware forward and turning actions, while stopping, backward motion, lateral translation, and ambiguous trajectories form a small set of specialized classes.

B Feature representations and Rarity

Feature representations and clustering dimensionality

We first construct a unified sparse representation from the aligned perception annotation after labeling by Grounding-DINO and Qwen3.6VL. After label normalization, each sample is encoded by 5 atomic features: `obj` denotes an observed object category; `count` denotes category multiplicity of at least two, capped at four; `loc` and `depth` attach the category to its image-plane and depth positions; `co` denotes an unordered pair of co-occurring categories, capped at 32 pairs per sample. For `loc` and `depth`, image-plane position is discretized into five horizontal regions from left to right, while vertical and depth information are represented by five additional regions. These atomic features form the common input vocabulary from which the context representation and joint perception-motion representation are derived.

The joint representation is then constructed by crossing the relevant atomic perception components with the rule-based meta-action labels. `joint_obj_motion` combines each `obj` category with the canonical `meta_action` and each of the four geometry tokens. `joint_slot_geometry` combines the same category together with its `loc/depth` slot and each geometry token. The `count` and `co` remain perception context features rather than entering this cross-

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product, preventing excessive vocabulary growth. Across all aligned samples, the resulting vocabulary contains 32,662 observed dimensions: 4,249 object–motion, 22,271 slot–geometry, 185 count, and 5,957 co-occurrence tokens. These dimensions correspond to the observed token vocabulary after sample alignment rather than a predefined dense feature length. Dimensions denote observed vocabulary sizes after sample alignment, rather than a predefined dense vector length. Rarity is measured by the length-normalized sum of token-level inverse document frequency (IDF), and the highest-scoring 5% of samples are selected as rare cases.

Here, we add the coverage chart of Sec. 5 in the main manuscript as well as the basis for the threshold selection. As illustrated in Fig. 1, the majority of episodes belong to the common–easy region, while rare–hard episodes represent only a small fraction of the benchmark but correspond to cases that are simultaneously underrepresented and challenging for the navigation policy. The hard and rare subsets are selected from their complete descending score distributions, independently of the subsequent reflection labels. For hard mining, the top 3% corresponds to 14,160 unique hard samples, with cutoffs of 0.270 m for ADE and 0.242 m for FDE. For rarity mining, the top 5% samples at a score cutoff of 34.94. In both rank curves, the selected boundary lies beyond the steepest high-score head and before the broad low-score body. The fixed 3% and 5% ratios therefore provide transparent tail-focused operating points while retaining enough samples for cluster- and reflection-level analysis; they are distribution-supported selection rules rather than thresholds optimized on downstream failure labels. The two tails capture related but non-equivalent phenomena. Thus, underrepresented perception–motion combinations account for a substantial portion of difficult predictions, but most difficult cases are not distributionally rare and most rare cases do not cause large trajectory errors. This partial overlap is consistent with the residual domain gap reported after real-world fine-tuning: a smaller real-world training set can improve average trajectory accuracy while still leaving uncommon visual–motion combinations insufficiently covered. The overlap is associative evidence and does not by itself establish rarity as the cause of prediction error.

C Reflection-Based Failure Attribution

Reflection Protocol

We analyze samples from the fixed hard set. For each sample, a teacher receives the student CoT and predicted trajectory together with historical frames, the current frame, ego-history, and the navigation goal. Privileged future frames and the ground-truth future trajectory are available only for verification. The teacher returns a schema containing five analysis fields (prediction, navigation goal, feasible area, rules, and collision risks), a planning rationale, and analysis/planning-/trajectory corrections. Failure labels are derived deterministically from this schema after reflection. The action-mismatch label is computed separately by comparing predicted and reference meta-actions. Labels are multi-label: a sample can simultaneously exhibit, for example, prediction, collision-risk, planning, and trajectory-shape errors. A mutually exclusive

primary label is retained only for summary; its fixed priority is Action > Predict > Goal > Feasible > Rule > Risk > Plan > Shape > Unexplained > Uncertain. It should not be used to infer the absence of secondary failures.

Figure 2 reports the multi-label incidence. `meta_action_mismatch` is computed after reflection by comparing the predicted action with the withheld reference meta-action. `trajectory_shape_error` is assigned when the teacher provides a non-empty trajectory correction. `unexplained_hard_sample` denotes a schema-valid reflection with no explicit failure signal despite membership in the fixed hard set; it is a coverage limitation, not evidence that the prediction is correct.

Qualitative cluster-conditioned cases

Here, we provide additional details of the clustering experiment in Sec.5 of the main manuscript. SVD reduces the raw 3597 tokens to 80 dimensions (51.96% explained variance), L2 normalization retains 80 dimensions, and a cosine UMAP maps them to a 20-dimensional space. This 20D space is further projected to 2D only for visualization, which does not determine cluster assignments. Figures 3–8 each contain two samples from one highlighted perception-context cluster. Every selected case is a non-noise member of the named cluster and contains the displayed bubble-aligned failure label. The panels show temporal visual evidence, the ego-centric history/ground-truth/predicted trajectories, the student prediction, and the complete teacher reflection. The selection covers at least two examples for Action, Rule, Predict, Risk, Plan, and Shape. It is intended to illustrate the localized associations in the enrichment analysis, not to estimate their prevalence.

The supplement supports a scene-conditioned interpretation: dense-pedestrian stopping contexts are associated with planning and trajectory-shape failures; scooter/sign and traffic-control contexts with action or rule failures; and scooter/cyclist interactions with prediction or collision-risk failures. The added hard–rare analysis shows that only a subset of these failure contexts is also underrepresented in the joint perception–motion distribution. These are statistically supported associations between independently generated reflection labels, fixed hard membership, and independently computed rarity; they are not exhaustive causal categories, particularly because a substantial portion of the fixed hard set has no explicit teacher-identified issue.

D Additional Experimental Results

Details of Real-world Deployment

We conducted two representative real-world experiments outdoors: street-side navigation involving an intersection crossing, and long-distance navigation within a university campus. The platform used was an AgileX Scout 2.0 mobile robot. All computation ran on a laptop equipped with an NVIDIA RTX 4090 GPU (16 GB), using only an Intel RealSense D455 camera as the perceptual input. The model achieved an average inference time of approximately 0.76 seconds. For navigation, we pre-sampled target points every 4 m on a pre-built point cloud map and sequentially fed these

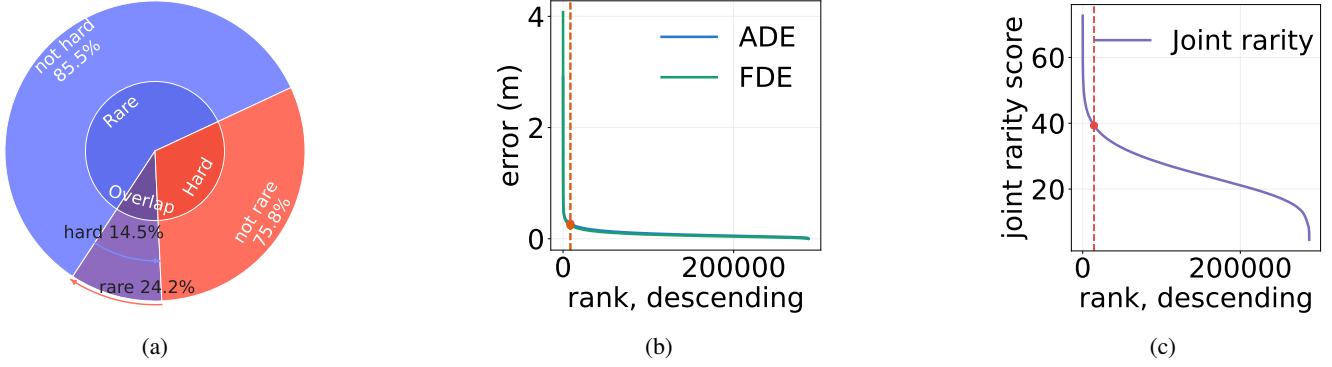


Figure 1: Each episode is categorized according to whether it belongs to the top-5% rarity subset and the hard subset identified by ADE/FDE ranking. The rare–hard region represents navigation cases that are both underrepresented in the dataset and difficult for the reference policy. (a) Overlap between the independently selected hard and rare subsets. (b) Descending ADE and FDE ranks. (c) Descending joint-rarity scores.

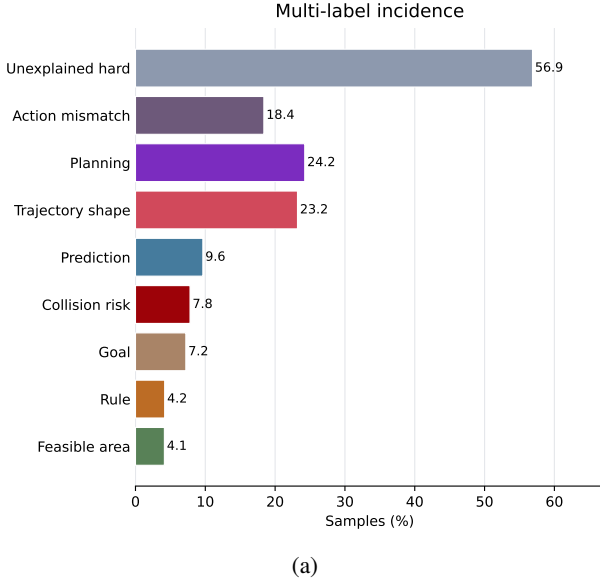


Figure 2: Multi-label distribution for failure distribution. The multi-label panel exposes co-occurring planning and trajectory failures that are necessarily suppressed when one primary label is selected.

points as navigation goals to the model; the point cloud map itself was not exploited by the online navigation algorithm. Given the current camera image and the target point, the model inferred a local trajectory for the next 5 seconds. A PD controller then drove the robot to track the first waypoint on this planned trajectory in real time, thereby closing the control loop. More detailed results are provided in the supplementary video, which simultaneously shows three views: the camera’s first-person view, a third-person view, and the LiDAR localization trajectory recorded by the FAST-LIO algorithm.

Hyperparameters for Model and Training.

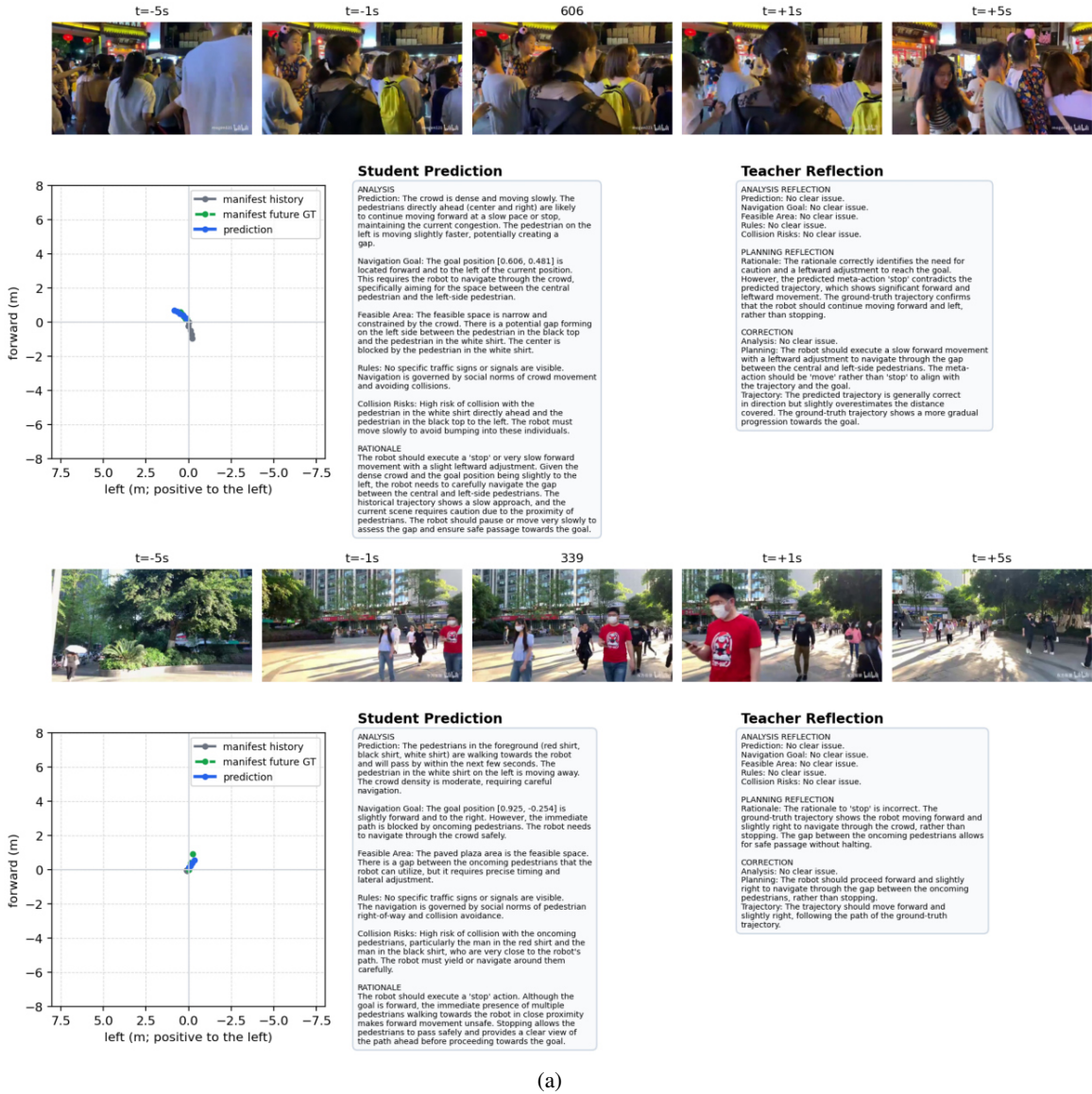
We fine-tune Qwen3.5-VL-4B (Wild-Nav) for urban navigation. The vision encoder is frozen; the LLM, vision–language merger, and trajectory head are trained. Inputs are 5-frame clips (640×360 , current + 4 history frames at 1 Hz) with dynamic resolution (100,352–602,112 pixels per frame), without fixed center cropping. The model predicts 5 future 2D waypoints via probe token `<PLAN_QUERY>`. Training runs for 1 epoch ($\sim 7.2k$ steps) on 8 GPUs (Nvidia RTX A6000) with DeepSpeed ZeRO-2, effective batch size 64, AdamW (LR 2×10^{-5} for backbone, 5×10^{-4} for action head), cosine schedule with 3% warmup, bf16, and gradient checkpointing.

Qualitative Results of Planning on Videos.

Figure 9, 10, and 11 below show the comparison of model prediction results. To evaluate the navigation capabilities of the models in fundamental environments, Figure 9 presents a qualitative comparison across standard, structured scenarios. These scenes encompass open park paths, well-structured commercial pedestrian streets, and relatively clear sidewalks. As observed from the visualizations, in these simple environments with sparse obstacles and clear spatial layouts, our reasoning-aware model (wildnav-thinking), the action-only variant (wildnav-inst), and the baseline (UrbanNav) all demonstrate high robustness. The trajectories planned by all three methods closely overlap with the human demonstration across all directions. This indicates that in standard scenarios lacking complex dynamic interactions, both existing multimodal navigation baselines and our proposed methods can accurately comprehend basic geometric constraints and smoothly navigate toward the goal.

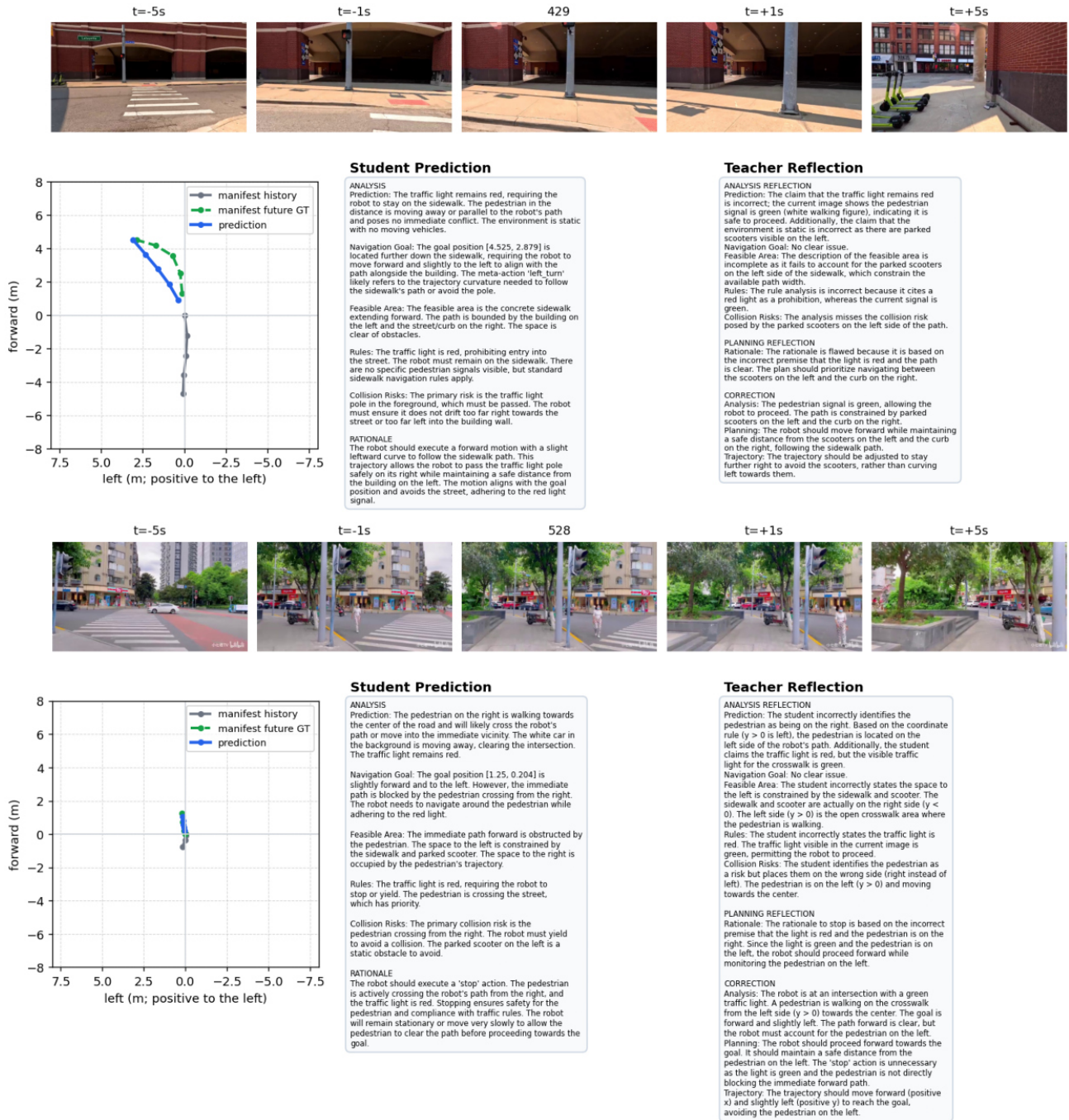
Figure 10. Qualitative comparison demonstrating the superiority of our methods over the UrbanNav baseline. Across various complex and highly constrained urban environments, our proposed methods consistently maintain accurate trajectory planning that aligns closely with the GT. In stark contrast, UrbanNav fails to adapt to these challenging scenes, exhibiting significant and potentially unsafe deviations from the target path.

To further investigate the critical role of structured CoT reasoning in complex environmental planning, Figure 11 compares the performance of wildnav-thinking against the action-only variant wildnav-inst in dynamic interaction scenarios, alongside visualizations of the intermediate reasoning text generated by the model. When facing highly challenging situations involving complex pedestrian intentions (e.g., a pedestrian with an umbrella, someone crossing a parking lot, or crowds near a barrier) and dynamic vehicles (e.g., a moving black van), the direct action mapping of wildnav-inst often struggles to fully capture the implicit risks in the scene. This limitation results in significant trajectory deviations and unreasonable path curvatures. In contrast, wildnav-thinking demonstrates a substantial advantage. By explicitly executing a two-stage "Analysis-Planning" reasoning process, the model accurately anticipates the future states of key obstacles (Prediction), delineates safe navigable spaces (Feasible area), and carefully evaluates potential hazards (Collision risks). Guided by this reasoning context, the action expert outputs meta-actions that are highly appropriate for the scene, thereby planning a safer, more reasonable, and robust trajectory that closely aligns with the ground truth (GT). This result intuitively demonstrates that incorporating high-level semantic reasoning into low-level trajectory planning is an effective approach to addressing long-tail dynamic interactions in complex urban environments.



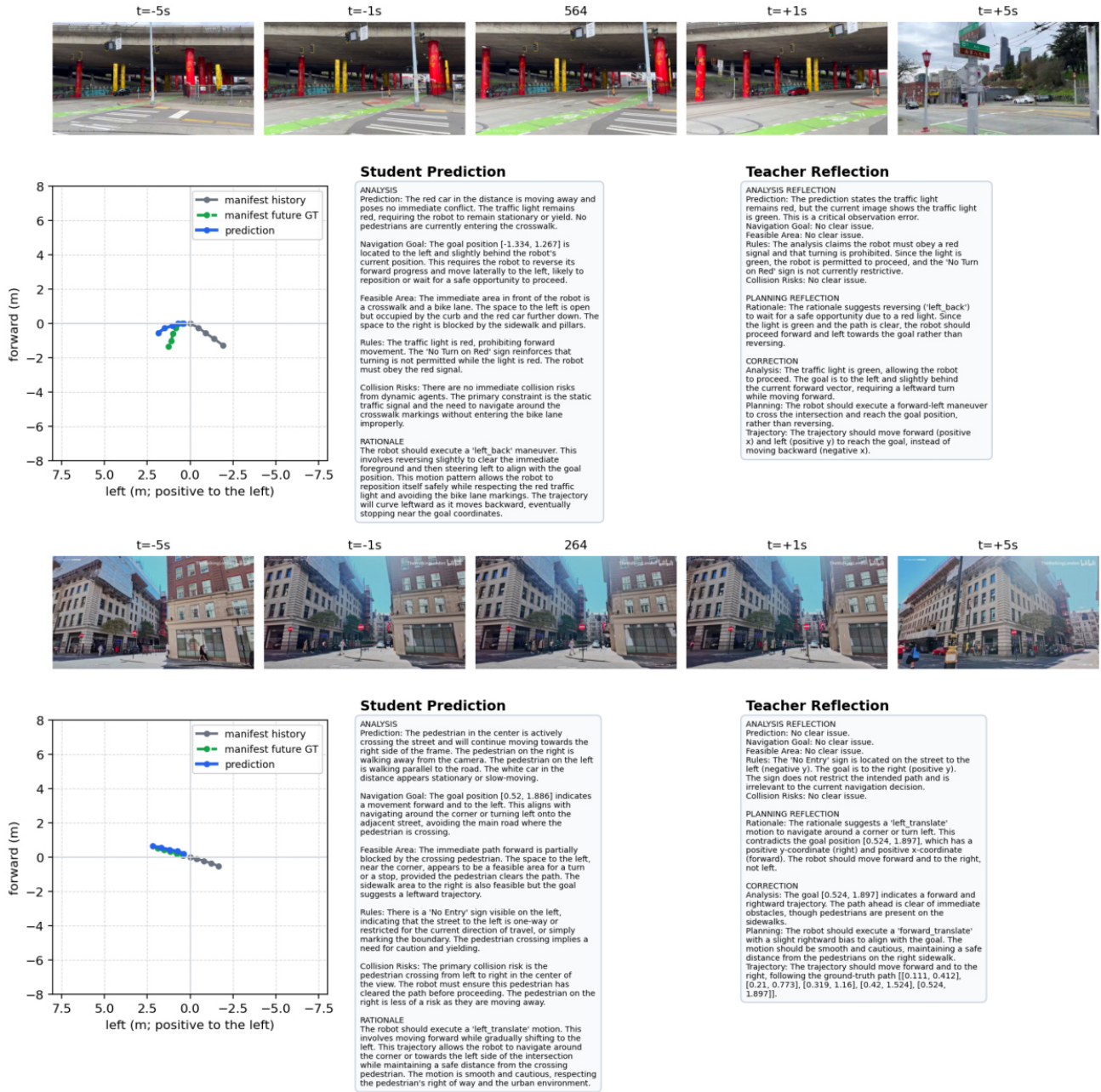
(a)

Figure 3: Dense-pedestrian stopping cases in C18. Both cases show planning and trajectory-shape errors: the student selects or justifies stopping although the verified trajectory permits controlled forward progress.



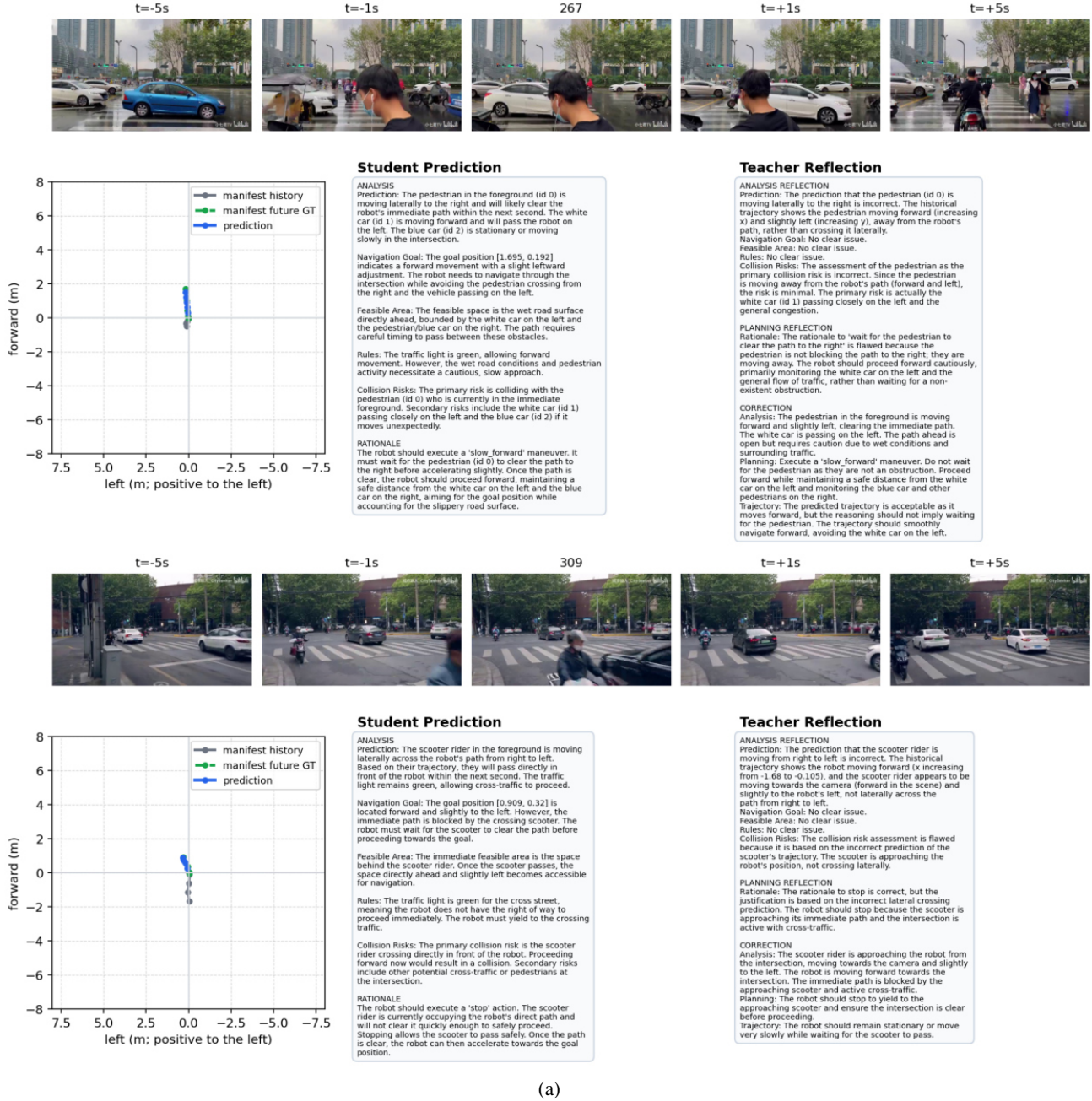
(a)

Figure 4: Scooter-sign crossing cases in C47. The cases combine action mismatch with rule interpretation errors around traffic signals and sidewalk versus vehicle-lane semantics.



(a)

Figure 5: Traffic-sign context cases in C60. The teacher identifies rule errors caused by treating signs or signals outside the intended pedestrian path as binding constraints, or by misreading their current state.



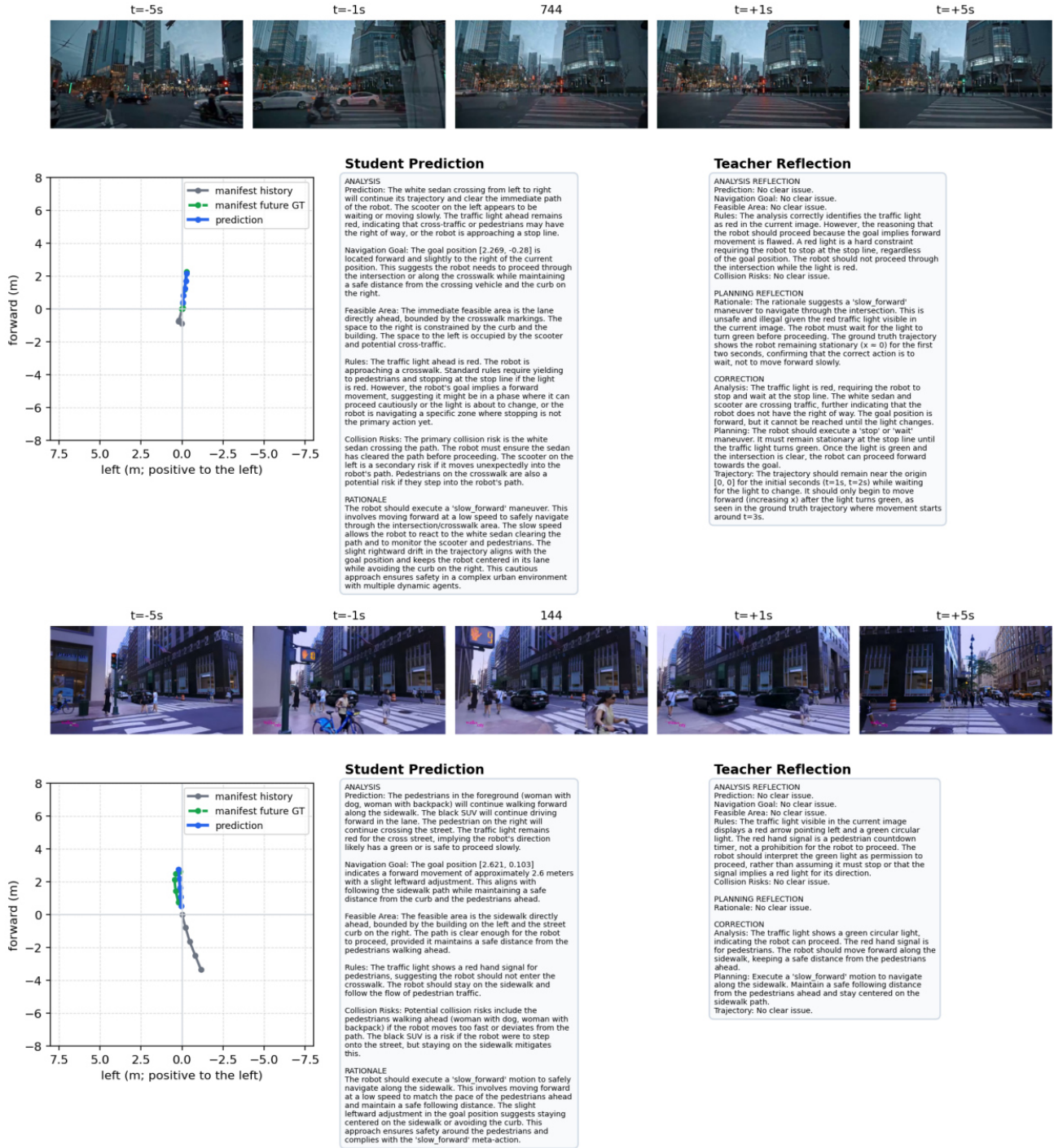
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Figure 6: Scooter-signal interaction cases in C105. The student mispredicts the motion or relevance of dynamic pedestrians and scooters, leading to incorrect near-term trajectory reasoning.



(a)

Figure 7: Cyclist-interaction cases in C141. The examples show under- or over-estimated collision risks caused by incorrect motion interpretation of cyclists, pedestrians, or cross-traffic.



(a)

Figure 8: Crosswalk-control cases in C155. The cases illustrate traffic-control reasoning errors, including incorrectly treating a signal as a hard constraint or misinterpreting pedestrian signal semantics.

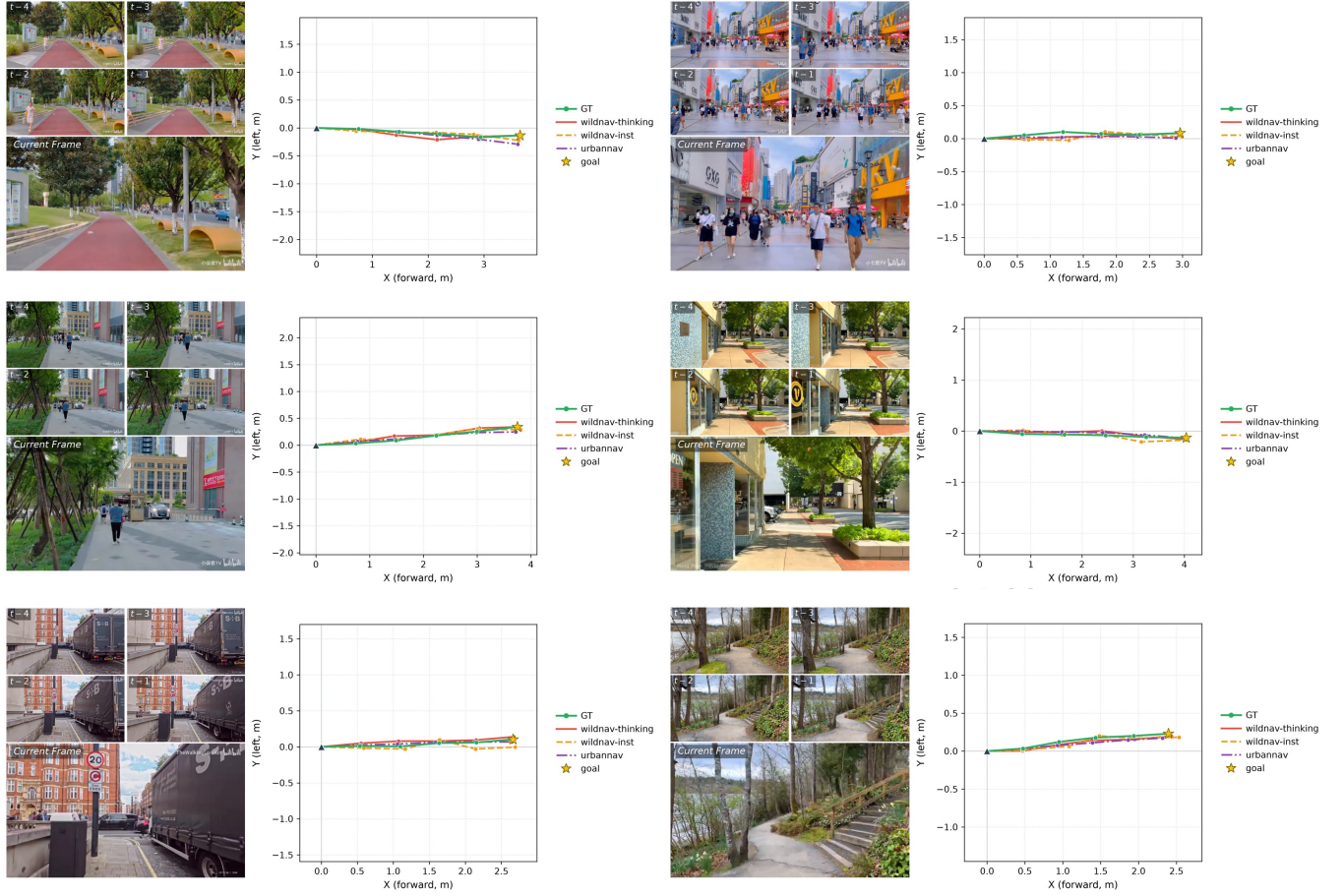


Figure 9: Qualitative comparison in standard navigation scenarios. In these relatively structured environments, both our proposed methods and the baseline (UrbanNav) demonstrate robust performance, generating trajectories that closely align with the ground truth (GT).

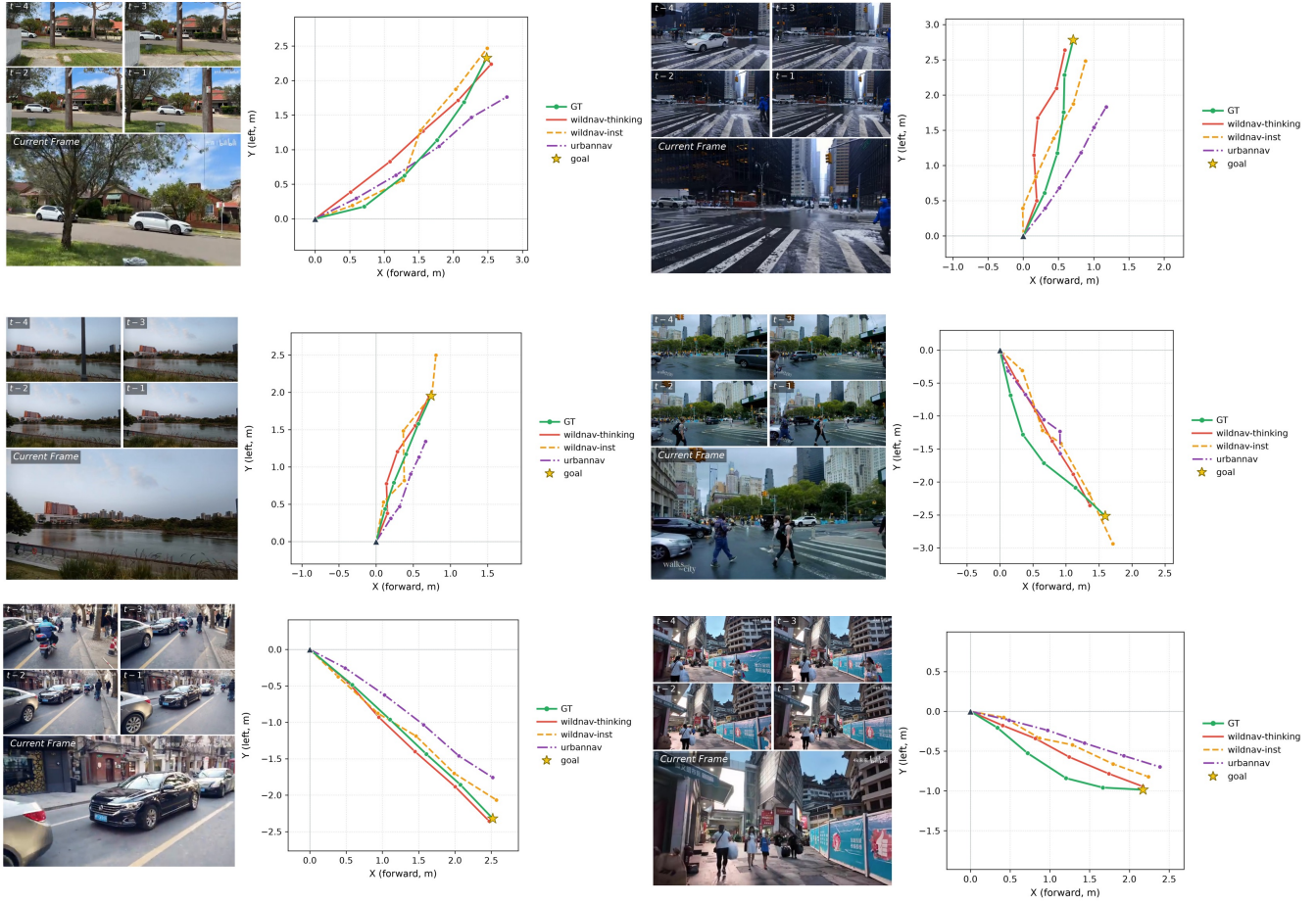
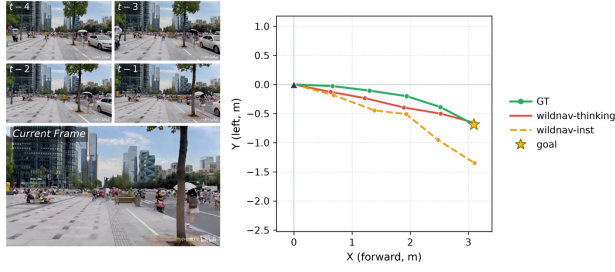


Figure 10: Qualitative comparison across diverse urban environments. Across various challenging scenarios, our proposed methods maintain accurate trajectory planning that aligns closely with the ground truth (GT). In contrast, the baseline (UrbanNav) exhibits significant deviations from the target path.

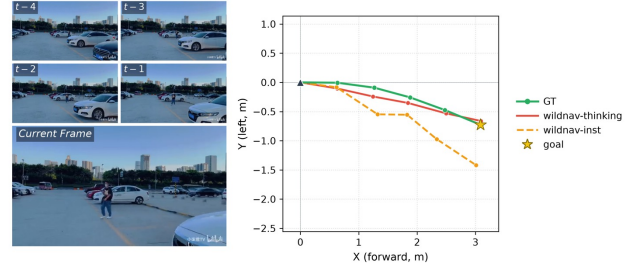


CoT (wildnav-thinking)

Analysis
 Prediction: The pedestrian in the yellow shirt will continue walking forward away from the robot. The pedestrian with the umbrella on the right appears stationary or moving slowly near the curb. The white car on the right is likely stationary or moving slowly in traffic. The general flow of pedestrians is moving away from the camera.
 Navigation goal: The goal position (3.093, -0.688) indicates a need to move forward and slightly to the right. This aligns with the 'slow_right_forward' meta-action, aiming to navigate past the immediate obstacles on the right (tree, pedestrian with umbrella) while maintaining a safe distance from the center path.
 Feasible area: The paved sidewalk provides a wide feasible area. The immediate path forward is clear, though the right side is constrained by the tree and the pedestrian with the umbrella. The left side is open but occupied by other pedestrians further ahead.
 Rules: No specific traffic signs or signals are visible that directly impact the immediate navigation on the sidewalk. Standard pedestrian right-of-way rules apply.
 Collision risks: The primary collision risk is the pedestrian with the umbrella on the right, who is close to the robot's intended path. The tree is a static obstacle to avoid. The pedestrian in the yellow shirt is moving away, posing less risk but requiring monitoring.

Planning

Meta-action: slow_right_forward
 Rationale: The robot should execute a 'slow_right_forward' motion. It needs to move forward to progress towards the goal while steering slightly right to align with the target position. The speed should be slow to safely pass the pedestrian with the umbrella on the right and navigate around the tree. The trajectory should maintain a safe lateral distance from the pedestrian with the umbrella and the tree, while staying on the sidewalk.

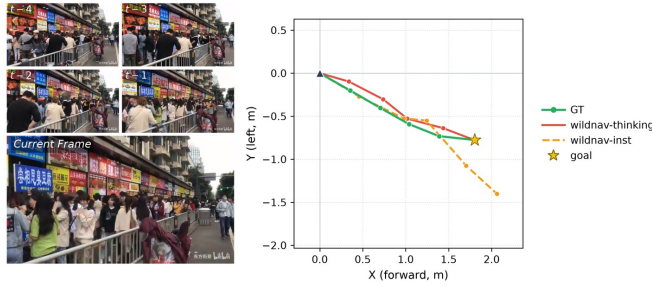


CoT (wildnav-thinking)

Analysis
 Prediction: The pedestrian in the center is walking towards the right side of the frame. Based on their trajectory, they will cross the robot's potential path and move towards the area near the white sedan. The parked cars remain stationary.
 Navigation goal: The goal position (3.086, -0.726) indicates a movement forward and slightly to the right. This aligns with the 'slow_right_forward' meta-action, aiming to navigate through the parking lot lane while maintaining a safe distance from the parked cars on the right.
 Feasible area: The feasible area is the open paved lane between the rows of parked cars. The path ahead is clear of immediate obstacles, though the pedestrian is a dynamic element to monitor.
 Rules: No specific traffic signs or signals are visible that dictate immediate movement rules, but standard parking lot etiquette applies (yielding to pedestrians).
 Collision risks: The primary risk is the pedestrian crossing the path. The robot must ensure the pedestrian has cleared the immediate trajectory before proceeding. The parked cars on the right define the boundary of the drivable space.

Planning

Meta-action: slow_right_forward
 Rationale: The robot should execute a 'slow_right_forward' motion. It will move forward while veering slightly right to align with the goal position. The speed should be low to account for the pedestrian's presence and to ensure a smooth, safe navigation through the parking lot lane. The trajectory will pass the pedestrian safely as they move to the right.

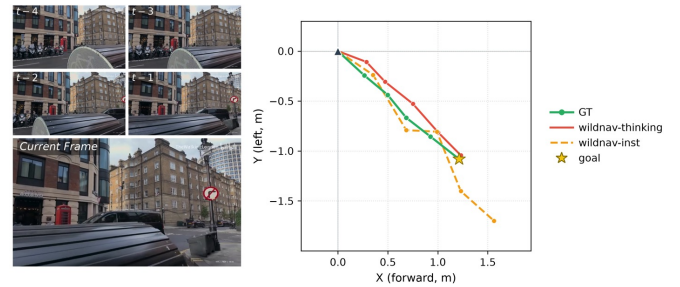


CoT (wildnav-thinking)

Analysis
 Prediction: The crowd on the left remains stationary behind the barrier. The pedestrian in the white jacket on the right is approaching the robot's position, while the pedestrian in the blue jacket continues moving away. The path ahead is relatively clear but requires careful navigation between the oncoming pedestrian and the barrier.
 Navigation goal: The goal position (1.803, -0.776) indicates a need to move forward and slightly to the right. This aligns with the 'slow_right_forward' meta-action, aiming to pass the oncoming pedestrian safely while maintaining distance from the barrier on the left.
 Feasible area: The feasible navigation space is the paved walkway to the right of the metal barrier. The path is narrow due to the barrier on the left and the presence of pedestrians on the right.
 Rules: No specific traffic signs or signals are visible. The implicit rule is to yield to oncoming pedestrians and maintain a safe distance from the crowd barrier.
 Collision risks: The primary risk is colliding with the oncoming pedestrian in the white jacket on the right. Secondary risks include bumping into the metal barrier on the left or the parked scooter further ahead.

Planning

Meta-action: slow_right_forward
 Rationale: The robot should execute a 'slow_right_forward' motion. It needs to move forward to progress towards the goal while steering slightly right to avoid the oncoming pedestrian in the white jacket and to center itself in the available walkway space. The speed should be low to react to the dynamic pedestrian environment.



CoT (wildnav-thinking)

Analysis
 Prediction: The black van on the right is moving forward and will continue to occupy the right side of the path. The pedestrian on the left is walking away and poses no immediate conflict. The path ahead is clear of static obstacles but constrained by the moving vehicle.
 Navigation goal: The goal position (1.212, -1.08) indicates a movement forward and to the right. This aligns with the 'slow_right_turn' meta-action, suggesting the robot needs to navigate around the immediate foreground obstacle (the white structure) and follow the path alongside the black van.
 Feasible area: The feasible space is the paved area between the parked motorcycles on the left and the moving black van on the right. The robot must stay within this corridor.
 Rules: A 'No Right Turn' sign is visible, but the robot's path is a straight/curved continuation along the street, not a turn into a side street, so it does not restrict the current motion.
 Collision risks: The primary risk is the black van on the right. The robot must maintain a safe lateral distance to avoid collision as it moves forward and slightly right. The white structure in the foreground also requires careful clearance.

Planning

Meta-action: slow_right_turn
 Rationale: The robot should execute a slow right turn to navigate around the white structure in the foreground and align with the path alongside the black van. The motion should be cautious due to the proximity of the van and the pedestrian on the left. The trajectory will curve gently to the right while moving forward, maintaining a safe distance from the van.

Figure 11: Visualization and evaluation of CoT reasoning. By explicitly generating a structured reasoning rationale (Analysis-Planning), our reasoning-aware model (wildnav-thinking) plans significantly more accurate and robust trajectories than its action-only counterpart (wildnav-inst).